

Adaptive Formation Tracking Control of Multiple Vertical Takeoff and Landing UAVs With Bearing-Only Measurements

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Abstract—This article studies the leader-following formation tracking control problem of multiple vertical takeoff and landing (VTOL) unmanned aerial vehicles (UAVs) subject to uncertain parameters, in which the target formation configuration is defined by using the interneighbors' bearing vectors. An adaptive formation control algorithm with bearing-only measurements is proposed under a hierarchical control framework. More specifically, a saturated adaptive distributed control force is developed in the position loop with bearing-only measurements. Since the bearing vectors with respect to the neighbors can be directly measured by low-cost onboard cameras, the proposed formation algorithm can be implemented without interagent communication. Subsequently, in the attitude loop, an adaptive hybrid control scheme via two modified Rodrigues parameters (MRPs) sets is proposed, which achieves the command attitude tracking globally and also avoids the unwinding problem of MRPs. Based on the Lyapunov stability analysis and hybrid theory, we prove that the overall closed-loop error system is globally asymptotically stable. Finally, we provide a numerical example to demonstrate the effectiveness of the proposed control algorithm.

Index Terms—Adaptive control, attitude control, bearing-only measurements, formation control, unmanned aerial vehicles (UAVs).

I. INTRODUCTION

FORMATION control of multiple unmanned aerial vehicles (UAVs) has drawn considerable attention due to its potential applications in military and civil fields [1], [2], [3]. As a typical class of UAVs, vertical takeoff and landing

(VTOL) is attractive due to its advantages in hovering and agile motion [4]. In contrast with the traditional single monolithic one, the coordinated formation of multiple VTOL UAVs is more feasible because even if some UAVs cannot work, the others still can reform a new formation and complete the mission. Note that the dynamical model of VTOL UAV is underactuated, in which four degree-of-freedom (DOF) control inputs require achieving 6-DOF motion including 3-DOF translation and 3-DOF rotational motion [4]. To deal with the underactuated issues, a hierarchical strategy is widely used in the control design, which consists of two loops, that is: 1) attitude control loop and 2) position control loop. Several typical control approaches, including adaptive control [5], model predictive control [6], immersion and invariance [7], backstepping control [8], and finite-time control [9], [10], have been proposed under a hierarchical framework.

In the viewpoint of the control structure, most formation control algorithms for VTOL UAVs are divided into centralized and distributed control. In particular, the centralized control (e.g., behaved-based control [11] and virtual structure control [12]) is effective for a small number of UAVs, but may fail in the case of large-scale formation since the large computational capability and communication bandwidth are required for the central node. Moreover, once the central node fails, the whole formation control will not be achieved. To overcome the drawbacks of the centralized one, distributed control approach has been developed, in which no central node is required and each node only needs information exchange with the neighbors. Inspired by the distributed idea, two formation control algorithms were proposed in [13] and [14] for multiple VTOL UAVs by considering communication delay and unavailable velocity information. In [15], a formation reconfiguration algorithm was developed for a group of VTOL UAVs. Dong et al. [16], Liu et al. [17], and Zou et al. [18] studied the formation control problems of VTOL UAVs under the switching topology. In addition, some distributed leader-following formation algorithms were proposed in [19], [20], and [21] for multiple VTOL UAVs.

Note that most formation control algorithms above are all displacement based, where the target formation is defined by the interagent displacement [22]. In contrast with the displacement-based one, bearing-based formation control algorithm, in which the interagent bearing vectors are used to define the target formation, owns more flexible formation maneuver capability since the bearing vectors are

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invariant to translational and scaling motions [23]. Inspired by this nice property, Zhao and Zelazo [23], Su et al. [24], Huang and Meng [25], and Li et al. [26] developed some bearing-based formation control algorithms to achieve the leader-following tracking. Different from bearing-based formation algorithms in [23], [24], [25], and [26] involving relative position or velocity information, bearing-only formation control that only relies on the bearing measurements with respect to the neighbors, is more challengeable since less measurement information is used. In [27], [28], [29], [30], [31], [32], and [33], several bearing-only formation control algorithms were proposed to only achieve a stationary target formation. An improved bearing-only formation algorithm was proposed in [34] to track a moving target formation. However, the algorithm of [34] requires the varying rate of the bearings. Trinh et al. [35] introduced the auxiliary systems and proposed the bearing-only formation algorithms without using the bearing's varying rates. Note that bearing-only formation algorithms in [34] and [35] for tracking moving targets are only applied to the single- and double-integrator multiagent systems (MASs). Few results consider the bearing-based formation problems of VTOL UAVs except the works in [25] and [26]. However, the results of [25] and [26] both require relative position and velocity of the neighbors.

In addition, most formation control algorithms of VTOL UAVs in the existing literature pay more attention in the position control loop and neglect some important issues in the attitude control loop. For instance, the Euler-angle-based attitude control law given in [36] did not consider the singularity problem. Quaternion-based attitude control laws in [13] and [14] neglected the unwinding problem. In particular, the unwinding problem means that a body (e.g., an UAV) rotates a large angle (more than π) instead of a small angle (less than π) to the desired attitude [37]. This leads to low control efficiency and high consumption of control energy. Even though the hysteresis-based switching attitude control law in [38] can solve the unwinding problem, the results of [38] require an extra binary variable and simultaneously rely on the precise model information.

This article focuses on the leader-following formation tracking control problem of multiple VTOL UAVs subject to uncertain parameters. An adaptive distributed formation control algorithm with bearing-only measurements is developed via a hierarchical framework. In particular, we first propose a distributed control force by resorting to adaptive technique. From the designed control force, we extract the command attitude as a reference tracking trajectory in the attitude loop. Then, an adaptive hybrid attitude control (HAC) law via two modified Rodrigue parameter (MRP) sets is developed. Compared with the related results, the main contributions of this article are threefold.

c1) In contrast with most UAV formation control algorithms in [13], [14], [15], [16], [17], and [18] that require relative position, velocity, or some extra auxiliary variables with respect to the neighbors, the proposed formation control algorithm relies on the bearing-only measurements that can be directly obtained by low-cost onboard cameras and, therefore, is easier to implement in practical applications.

c2) Unlike some existing bearing-only formation control algorithms in [27], [28], [29], [30], [31], [32], and [33] that only achieve a stationary target formation, the developed bearing-only formation algorithm achieves tracking a moving target formation for multiple VTOL UAVs dynamical systems with model uncertainties. By resorting to adaptive technique, the proposed algorithm does not require the precise model formation.

c3) We propose a new MRPs-based adaptive hybrid control law in the attitude loop. By switching on the two MRPs sets, the proposed algorithm avoids the singularity problem of MRPs and achieves global attitude tracking control objective, but also can solve the unwinding problem of MRPs.

The remainder of this article is organized as follows. Section II formulates the formation control problem of VTOL UAVs. Section III presents the main results including algorithm design and stability analysis. Section IV performs a simulation example and Section V draws some conclusions.

II. PROBLEM STATEMENT

Notation: $\mathbb{R}^{n \times n}$ and $\mathbb{R}^n$ denote the sets of $n \times n$ real matrices and n dimension real vectors. I_n is the $n \times n$ identity matrix and $\mathbf{1}_n$ is the $n \times 1$ ones vector. $|\cdot|$ is the absolute value of a scalar, and $\|\cdot\|$ is the Euclidean norm. Let $\text{col}(x_1, \dots, x_n)$ be a column stack with x_i over i from 1 to n , $\text{diag}(x_1, \dots, x_n)$ be a diagonal matrix and $\otimes$ denotes the Kronecker product. For any $x = [x_1, x_2, x_3]^T \in \mathbb{R}^3$, $x^\times = [0, -x_3, x_2; x_3, 0, -x_1; -x_2, x_1, 0]$. $\tanh(x) = (e^x - e^{-x})/(e^x + e^{-x}) \in (-1, 1)$ and $\cosh(x) = (1/2)(e^x + e^{-x}) \in [1, \infty)$ are the hyperbolic tangent and cosine functions, respectively. $\mathbb{L}_2$ and $\mathbb{L}_\infty$ are the function spaces, which are defined by $\mathbb{L}_2 = \{f : \mathbb{R}^+ \rightarrow \mathbb{R}^n \mid \int_0^\infty \|f(t)\|^2 dt < \infty\}$ and $\mathbb{L}_\infty = \{f : \mathbb{R}^+ \rightarrow \mathbb{R}^n \mid \text{esssup}_{t \in \mathbb{R}^+} \|f(t)\| < \infty\}$.

Consider a group of n VTOL UAVs, labeled by $\mathcal{V} = \{1, 2, \dots, n\}$. Each UAV is viewed as a rigid body and its 6-DOF motion is modeled in two reference frames: 1) the global inertia frame $\mathcal{F}_I = \{O_I x_I y_I z_I\}$, whose origin is located at the Earth center and 2) the body-fixed frame $\mathcal{F}_{bi} = \{O_{bi} x_{bi} y_{bi} z_{bi}\}$, whose origin is situated at the mass center of the i th UAV. The attitude of the i th UAV, which represents the rotation relationship between frames $\mathcal{F}_I$ and $\mathcal{F}_{bi}$, is parameterized by the MRPs $\sigma_i = \hat{k}_i \tan((\phi_i/4))$, where $\hat{k}_i \in \mathbb{S}_2 = \{x \in \mathbb{R}^3 \mid \|x\| = 1\}$ is a Euler axis and $\phi_i \in (-2\pi, 2\pi)$ is a Euler angle.

Based on the Euler-Newton formula, the 6-DOF motion of each UAV is formulated as

$$\dot{p}_i = v_i \quad (1a)$$

$$m_i \dot{v}_i = -m_i g \hat{e}_3 + T_i R(\sigma_i) \hat{e}_3 \quad (1b)$$

$$\dot{\sigma}_i = E(\sigma_i) \omega_i \quad (1c)$$

$$J_i \dot{\omega}_i = -\omega_i^\times J_i \omega_i + \tau_i \quad (1d)$$

where $p_i \in \mathbb{R}^3$ and $v_i \in \mathbb{R}^3$ are the i th UAV's position and velocity expressed in frame $\mathcal{F}_I$, m_i is its total mass, g denotes the gravity acceleration, $\hat{e}_3 = [0, 0, 1]^T$, T_i is the thrust magnitude along the orientation $\hat{e}_3$, σ_i is the MRPs to represent the i th UAV's attitude and $R(\sigma_i) = I_3 + 4(1 - \sigma_i^T \sigma_i / (1 + \sigma_i^T \sigma_i)^2) \sigma_i^\times + (8 \sigma_i^\times \sigma_i^\times / (1 + \sigma_i^T \sigma_i)^2)$ is the rotation matrix from frame $\mathcal{F}_{bi}$ to frame $\mathcal{F}_I$, $E(\sigma_i) =$

$(1/4)((1 - \sigma_i^T \sigma_i)I_3 + 2\sigma_i^\times + 2\sigma_i \sigma_i^T)$, ω_i is the angular velocity expressed in frame $\mathcal{F}_{bi}$, $J_i \in \mathbb{R}^{3 \times 3}$ is the inertia matrix with respect to frame $\mathcal{F}_{bi}$, and $\tau_i \in \mathbb{R}^3$ is the control torque expressed in frame $\mathcal{F}_{bi}$. The precise physical parameters including mass m_i and inertia matrix J_i are not available. In addition, some properties with respect to the MRPs are presented as follows [39]:

$$\sigma_i^T E(\sigma_i) = \left(\frac{1 + \sigma_i^T \sigma_i}{4} \right) \sigma_i^T \quad (2a)$$

$$E(\sigma_i) E^T(\sigma_i) = \left(\frac{1 + \sigma_i^T \sigma_i}{4} \right)^2 I_3. \quad (2b)$$

Assumption 1: The precise mass m_i of the i th UAV is not available while its upper bound $\bar{m}_i$ and lower bound $\underline{m}_i$ are known.

Remark 1: In contrast with rotation matrix and unit quaternion, the MRPs are more attractive since they provide a minimal parameter attitude representation and also allow a large rotation in the range $(-2\pi, 2\pi)$ except the singularity points $\phi_i = \pm 2\pi$. To avoid the singularity issue, a typical method is to introduce its shadow counterpart $s(\sigma_i) = -(\sigma_i / \sigma_i^T \sigma_i)$ [40]. By combining of these two MRPs sets, it is capable of achieving global and nonsingular attitude representation.

A formation framework of n VTOL UAVs is defined by $(\mathcal{G}, p)$, where $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ denotes the topology graph of n UAVs with the node set $\mathcal{V} = \{1, 2, \dots, n\}$ and the edge set $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$, and $p = \text{col}(p_1, p_2, \dots, p_n) \in \mathbb{R}^{3n}$ is a configuration of n UAVs. The node set $\mathcal{V}$ consists of the sets of followers and leaders, that is, $\mathcal{V}_f = \{1, \dots, n_f\}$ and $\mathcal{V}_l = \{n_f + 1, \dots, n\}$, where n_f and $n_l = n - n_f$ are the numbers of the followers and leaders, respectively. An edge $(i, j) \in \mathcal{E}$ in the graph $\mathcal{G}$ implies that UAV i can measure relative information (e.g., bearing information) with respect to the neighbors. $\mathcal{N}_i = \{j \mid (i, j) \in \mathcal{E}\}$ denotes all the neighbors of UAV i . An undirected graph $\mathcal{G}$ is considered in this article, and the bearing rigidity is employed to define the target formation. Given any two UAVs $i, j \in \mathcal{V}$, their relative position and bearing vectors are defined by

$$e_{ij} = p_j - p_i, g_{ij} = \frac{p_j - p_i}{\|p_j - p_i\|}, (i, j) \in \mathcal{E}.$$

Note that $e_{ij} = -e_{ji}$ and $g_{ij} = -g_{ji}$. For the bearing vector g_{ij} , $P_{g_{ij}} = I_3 - g_{ij}g_{ij}^T$ is an orthogonal projection matrix.

For an undirected graph $\mathcal{G}$, we can assign arbitrarily the orientation of the edges and index all the edges as $e_1, e_2, \dots, e_m$, where $m = |\mathcal{E}|$ is the total edge number. Then, the incidence matrix $H = [h_{ki}] \in \mathbb{R}^{m \times n}$ is used to describe the relationship between nodes and edges, in which the element h_{ki} is defined as $h_{ki} = -1$ if node i is the tail of edge k , $h_{ki} = 1$ if node i is the head of edge k , and $h_{ki} = 0$, otherwise. Let $e = \text{col}(e_1, \dots, e_m) \in \mathbb{R}^{3m}$, and one has that

$$e = (H \otimes I_3)p = \bar{H}p.$$

Given the desired bearing vectors $\{g_{ij}^*\}_{(i,j) \in \mathcal{E}}$, we define the target formation $(\mathcal{G}, p^*)$ as follows.

Definition 1: A bearing-constrained target formation $(\mathcal{G}, p^*)$ is defined by satisfying the following two conditions: 1) $((p_j^*(t) - p_i^*(t)) / \|p_j^*(t) - p_i^*(t)\|) = g_{ij}^* \forall (i, j) \in \mathcal{E}$ and 2) $p_i^*(t) = p_i(t) \forall i \in \mathcal{V}_l$.

The following assumption is imposed and its role is to ensure the uniqueness of target formation $(\mathcal{G}, p^*)$.

Assumption 2: The target formation framework $(\mathcal{G}, p^*)$ is infinitesimally bearing rigid and $n_l \geq 2$.

Based on the desired bearing vectors $\{g_{ij}^*\}_{(i,j) \in \mathcal{E}}$, we derive a bearing-based Laplacian matrix $L = [L_{ij}] \in \mathbb{R}^{3n \times 3n}$, where $L_{ii} = \sum_{j \in \mathcal{N}_i} P_{g_{ij}^*}$, $L_{ij} = -P_{g_{ij}^*}$ if $(i, j) \in \mathcal{E}$, and $L_{ij} = 0_{3 \times 3}$, otherwise. Moreover, the Laplacian matrix L can be decomposed into

$$L = \begin{bmatrix} L_{ff} & L_{fl} \\ L_{lf} & L_{ll} \end{bmatrix}$$

where $L_{ff} \in \mathbb{R}^{3n_f \times 3n_f}$, $L_{fl} \in \mathbb{R}^{3n_f \times 3n_l}$, and $L_{ll} \in \mathbb{R}^{3n_l \times 3n_l}$.

Define $p = \text{col}(p_f, p_l) \in \mathbb{R}^{3n}$ and $v = \text{col}(v_f, v_l) \in \mathbb{R}^{3n}$, where $p_f, v_f \in \mathbb{R}^{3n_f}$ are the position and velocity vectors of all the followers, and $p_l, v_l \in \mathbb{R}^{3n_l}$ are the position and velocity vectors of all the leaders. According to [23, Th. 1], one has that the target formation $(\mathcal{G}, p^*)$ is unique under Assumption 2. Moreover, the position and velocity vectors of all the followers in the target formation $(\mathcal{G}, p^*)$ are obtained by $p_f^* = -L_{ff}^{-1} L_{fl} p_l$ and $v_f^* = -L_{ff}^{-1} L_{fl} v_l$. In this article, we consider the case that $n_l = n - n_f$ leader UAVs move at a common constant velocity, that is, $v_l = \text{col}(v_{n_f+1}, \dots, v_n) = \mathbf{1}_{n_l} \otimes v_d$ with a constant vector $v_d \in \mathbb{R}^3$. Based on [23, Lemma 1], we obtain that $\ker(L) = \text{Im}(\mathbf{1}_n \otimes I_3, p^*)$. Since $v_l = \mathbf{1}_{n_l} \otimes v_d$, it implies that $v_f^* = \mathbf{1}_{n_f} \otimes v_d$.

This article aims to design the control thrust T_i and torque τ_i using bearing-only measurements with respect to the neighbors, that is, $\{g_{ij}\}_{j \in \mathcal{N}_i}$ such that the bearing-constrained target formation $(\mathcal{G}, p^*)$ can be achieved, that is

$$\lim_{t \rightarrow \infty} p_f(t) - p_f^* = 0, \lim_{t \rightarrow \infty} v_f(t) - v_f^* = 0. \quad (3)$$

III. MAIN RESULTS

In this section, we develop a distributed control algorithm with only relative bearing measurements to solve the considered leader-following formation tracking problem of multiple VTOL UAVs. To deal with the underactuated issue of VTOL UAVs system, a hierarchical control strategy is employed. In particular, we first propose an adaptive distributed control force for each UAV in the position loop to achieve the leader-following formation tracking. Then, we extract the thrust magnitude and command attitude from the proposed control force, and an MRPs-based adaptive hybrid control law is synthesized in the attitude loop such that global and anti-winding attitude tracking is achieved. Finally, we provide the stability analysis of the closed-loop system.

A. Problem Transformation

Let $p_f^* = \text{col}(p_{f,1}^*, \dots, p_{f,n_f}^*)$ and $v_f^* = \text{col}(v_{f,1}^*, \dots, v_{f,n_f}^*)$, where $p_{f,i}^*$ and $v_{f,i}^*$ are the desired position and velocity of the i th UVA in the target formation $(\mathcal{G}, p^*)$. For any $i \in \mathcal{V}_f$, define the position error $p_i^e = p_i - p_{f,i}^*$ and velocity error $v_i^e = v_i - v_{f,i}^*$.

According to the model (1), the dynamics of the errors p_i^e and v_i^e are given by

$$\dot{p}_i^e = v_i^e \quad (4a)$$

$$m_i \dot{v}_i^e = -m_i g e_3 + u_i + T_i (R(\sigma_i) - R(\sigma_i^c)) e_3 \quad (4b)$$

where $u_i = T_i R(\sigma_i^c) e_3$ is an intermediate control force to be designed later and $R(\sigma_i^c)$ is a rotational matrix with respect to the MRPs σ_i^c . Based on the determined control force u_i , we obtain the thrust magnitude T_i as

$$T_i = \|u_i\|, i \in \mathcal{V}_F \quad (5)$$

with the fact that $\|R(\sigma_i^c) e_3\| = 1$. Also, we extract the command attitude σ_i^c from the control force u_i , which can be obtained from [14, Lemma 1].

Lemma 1: Suppose that the control force $u_i = [u_{i,x}, u_{i,y}, u_{i,z}]^T$ satisfies the condition

$$u_i \notin \mathbb{U} = \left\{ u \in \mathbb{R}^3 \mid u = [0, 0, u_z]^T, u_z \leq 0 \right\}. \quad (6)$$

Then, the command attitude σ_i^c can be extracted by

$$\sigma_i^c = \frac{1}{2(1 + q_{i,0})T_i q_{i,0}} \begin{bmatrix} -u_{i,y} \\ u_{i,x} \\ 0 \end{bmatrix} \quad (7)$$

where $q_{i,0} = \sqrt{(1/2) + (u_{i,z}/2T_i)}$.

In fact, the condition (6) can be understood from two aspects. On the one hand, from the view of theoretical analysis, if the command force $u_i = T_i R(\sigma_i^c) e_3 \in \mathbb{U}$, it follows that u_i and e_i are collinear with opposite directions. It follows from (7) that $q_{i,0} = \sqrt{(1/2) + (u_{i,z}/2T_i)} = 0$ and σ_i^c encounters singularity at the point $q_{i,0} = 0$. Thus, to extract the command attitude without the singularity problem, the condition (6) should be satisfied. On the other hand, from the view of practical applications, it follows from the definition of the set $\mathbb{U}$ that if $u_i \in \mathbb{U}$, the orientation of command force u_i is the same as the gravitational acceleration. This situation should be avoided in practical applications.

Next, we define the attitude error between the MRPs σ_i and command attitude σ_i^c as [41]

$$\sigma_i^e = \frac{\sigma_i^c (\sigma_i^T \sigma_i - 1) + \sigma_i (1 - \|\sigma_i^c\|^2) - 2(\sigma_i^c)^{\times} \sigma_i}{1 + \|\sigma_i^c\|^2 \|\sigma_i\|^2 + 2(\sigma_i^c)^T \sigma_i}.$$

Note that $\sigma_i^e = [0, 0, 0]^T$ implies that the command attitude σ_i^c can be tracked, that is, $\sigma_i = \sigma_i^c$. Based on the works of [41], the dynamics of σ_i^e is given by

$$\dot{\sigma}_i^e = E(\sigma_i^e) \omega_i^e, \omega_i^e = \omega_i - R^T(\sigma_i^e) \omega_i^c \quad (8)$$

where ω_i^c is the command angular velocity. According to (1) and (8), the dynamics of ω_i^e is

$$J_i \dot{\omega}_i^e = -\omega_i^{\times} J_i \omega_i + \tau_i + J_i \left((\omega_i^e)^{\times} R^T(\sigma_i^e) \omega_i^c - R^T(\sigma_i^e) \dot{\omega}_i^c \right) \quad (9)$$

where ω_i^c and its derivative can be derived from the command attitude σ_i^c , and their detailed expressions can be seen in [14, Lemma 1].

We summarize the above problem transformation in the following lemma.

Lemma 2: Consider the error dynamical systems (4), (8), and (9). If the developed control force u_i and attitude control torque τ_i guarantee that

$$\lim_{t \rightarrow \infty} p_i^e(t) = 0, \lim_{t \rightarrow \infty} v_i^e(t) = 0 \quad (10)$$

$$\lim_{t \rightarrow \infty} \sigma_i^e(t) = 0, \lim_{t \rightarrow \infty} \omega_i^e(t) = 0 \quad (11)$$

for $\forall i \in \mathcal{V}_F$, the formation control objective (3) is achieved.

B. Distributed Control Force Design

In this section, a distributed control force u_i is developed for the error system (4) such that $\lim_{t \rightarrow \infty} p_i^e(t) = 0$, $\lim_{t \rightarrow \infty} v_i^e(t) = 0$ for $\forall i \in \mathcal{V}_F$. As will be shown in the following stability analysis, the boundedness of T_i is an important condition for the closed-loop system stability. To this end, we propose a distributed command control strategy with saturated property. Moreover, an adaptive control law is introduced to deal with the problem of unavailable mass parameter.

For any $i \in \mathcal{V}_F$, a saturated distributed control force is developed as

$$u_i = -k_s \tanh(\varpi_i) + k_p \sum_{j \in \mathcal{N}_i} (g_{ij} - g_{ij}^*) + \hat{m}_i (g \hat{e}_3 + \dot{\chi}_i) \quad (12)$$

where $\varpi_i = v_i - \chi_i$, k_s and k_p are positive constants, $\tanh(\cdot)$ is the hyperbolic tangent function, $\hat{m}_i$ is an estimation of the mass m_i , and χ_i is an output of the following auxiliary system:

$$\dot{\xi}_i = -k_1 \tanh(\xi_i - \chi_i) \quad (13a)$$

$$\dot{\chi}_i = k_2 k_p \sum_{j \in \mathcal{N}_i} (g_{ij} - g_{ij}^*) - k_3 \dot{\xi}_i \quad (13b)$$

with $k_1, k_2, k_3 > 0$. In addition, the adaptive updated law of $\hat{m}_i$ is designed as

$$\dot{\hat{m}}_i = \begin{cases} -\gamma_i (g \hat{e}_3 + \dot{\chi}_i)^T \varpi_i, & \text{if } \underline{m}_i < m_i < \bar{m}_i \\ & \text{or } \hat{m}_i = \bar{m}_i \text{ with } \psi_i < 0 \\ & \text{or } \hat{m}_i = \underline{m}_i \text{ with } \psi_i > 0 \\ 0, & \text{if } \hat{m}_i = \bar{m}_i \text{ with } \psi_i \geq 0 \\ & \text{or } \hat{m}_i = \underline{m}_i \text{ with } \psi_i \leq 0 \end{cases} \quad (14)$$

where $\psi_i = -(g \hat{e}_3 + \dot{\chi}_i)^T \varpi_i$, $\gamma_i > 0$, $\underline{m}_i$ and $\bar{m}_i$ are the upper and lower bounds of the mass m_i . According to the results of [42], the estimation state $\hat{m}_i$ can be always evolved in the interval $[\underline{m}_i, \bar{m}_i]$ if its initial value is chosen in this range. In addition, since $\hat{m}_i(t) \in [\underline{m}_i, \bar{m}_i]$ for $\forall t \geq 0$, it follows from (5) and (12) that the thrust magnitude T_i satisfies that:

$$T_i = \|u_i\| \leq \sqrt{3} k_c + \bar{m}_i (g + \sqrt{3} k_o). \quad (15)$$

In general, some relative information (e.g., relative position or velocity) with respect to the neighbors is required for each UAV to achieve the formation control objective. For instance, the formation controllers of [15], [16], and [17] involve the relative position and velocity with respect to the neighbors. Besides the relative position and velocity, the controllers of [13] and [14] also require relative information of some auxiliary variables. Note that the relative position and velocity of [15], [16], and [17] can be measured by some

onboard sensors or obtained by interagent communication. However, the relative auxiliary variables of [13] and [14] cannot be measured directly and are only obtained by interagent communication. From (12)–(14), we know that the proposed distributed control force only involves the bearing vectors $\{g_{ij}\}_{j \in \mathcal{N}_i}$, which can be measured by low-cost onboard cameras mounted on the UAV without interagent communication. Thus, compared with the controllers of [13], [14], [15], [16], and [17], the formation control law (12) is more easier to implement in practical applications.

Remark 2: The distributed control force (12) is partly motivated by the results of [34] and [35], in which two bearing-based formation algorithms were proposed for single- and double-integrator MASs. We extend the formation control algorithms of [34] and [35] to the complex underactuated and nonlinear UAV dynamic systems with uncertain parameters. Moreover, based on projection-based adaptive technique, the thrust magnitude of the algorithm (12) is shown to be bounded and, therefore, can be used to solve the thrust saturation issue of UAVs.

C. Attitude Control Design

In this section, we focus on designing a global attitude tracking controller for the error systems (8), (9) such that $\lim_{t \rightarrow \infty} \sigma_i^e(t) = 0$, $\lim_{t \rightarrow \infty} \omega_i^e(t) = 0$ for any $\sigma_i^e(0), \omega_i^e(0)$.

Since the MRPs $\sigma_i = \hat{k}_i \tanh([\phi_i/4])$ encounter the singularity problem at $\phi_i = \pm 2\pi$, many MRPs-based attitude controllers (e.g., [41]) only achieve local or almost global stability. In addition, similar to unit quaternion, the MRPs-based controller may suffer from the unwinding problem. To achieve global stability and avoid the unwinding problem, O'Keefe and Schaub [40] and Schaub et al. [43] introduced the shadow counterpart $s(\sigma_i)$ and proposed the discontinuous strategies by switching between the MRPs σ_i and its shadow counterpart $s(\sigma_i)$ at the surface $\|\sigma_i\| = 1$. However, such a switching strategy at $\|\sigma_i\| = 1$ is very sensitive to perturbations and measurement noises. To deal with this issue, inspired by the hybrid control method in [38], we propose a new attitude tracking strategy to achieve global stability and simultaneously avoid the unwinding problem. Moreover, an adaptive control law is incorporated to address the problem of unavailable inertia matrix.

First, define the inertia matrix J_i as

$$J_i = \begin{bmatrix} J_{i,11} & J_{i,12} & J_{i,13} \\ J_{i,12} & J_{i,22} & J_{i,23} \\ J_{i,13} & J_{i,23} & J_{i,33} \end{bmatrix}.$$

For any vector $a = [a_1, a_2, a_3]^T$, a linear operator $\Upsilon(a)$ is introduced by

$$\Upsilon(a) = \begin{bmatrix} a_1 & 0 & 0 & 0 & a_3 & a_2 \\ 0 & a_2 & 0 & a_3 & 0 & a_1 \\ 0 & 0 & a_3 & a_2 & a_1 & 0 \end{bmatrix}.$$

By using the linear operator $\Upsilon(a)$, (9) is rewritten as

$$J_i \dot{\omega}_i^e = -\omega_i^{\times} \Upsilon(\omega_i) \theta_i + \tau_i + \Upsilon(\Lambda_i) \theta_i \quad (16)$$

where $\Lambda_i = (\omega_i^e)^{\times} R^T(\sigma_i^e) \omega_i^c - R^T(\sigma_i^e) \dot{\omega}_i^c$ and $\theta_i = [J_{i,11}, J_{i,22}, J_{i,33}, J_{i,23}, J_{i,13}, J_{i,12}]^T$ are the unknown parameters in the inertia matrix J_i .

Defining $x_i = [\sigma_i^e, \omega_i^e]^T$, and an adaptive hybrid attitude tracking control law for the error systems (8) and (9) is proposed as

$$\tau_i(x_i) = -k_\omega(\omega_i^e - \zeta_i) - k_\sigma \sigma_i^e - \left(\Upsilon(\Lambda_i - \dot{\zeta}_i) - \omega_i^{\times} \Upsilon(\omega_i) \right) \hat{\theta}_i, x_i \in \mathcal{C}_i \quad (17a)$$

$$x_i^+ = G(x_i) = (s(\sigma_i^e), \omega_i^e), x_i \in \mathcal{D}_i \quad (17b)$$

where k_ω and k_σ are positive constants, $\hat{\theta}_i$ is an estimation of the unknown parameter θ_i , and ζ_i is the auxiliary variable generated by

$$\dot{\zeta}_i = -l_1 \zeta_i - l_2 k_\sigma \sigma_i^e \quad (18)$$

where l_1 and l_2 are positive constants. In addition, $\mathcal{C}_i$ and $\mathcal{D}_i$ of (17) are the flow set and jump set, which are defined as

$$\mathcal{C}_i = \{x_i \in \mathbb{R}^6 : \|\sigma_i^e\| \leq 1 + \delta_i\}$$

$$\mathcal{D}_i = \{x_i \in \mathbb{R}^6 : \|\sigma_i^e\| \geq 1 + \delta_i\}$$

where $\delta_i > 0$ is called a hysteresis constant. From the definition of the flow set $\mathcal{C}_i$ and jump set $\mathcal{D}_i$, it follows that if $\|\sigma_i^e\| \geq 1 + \delta_i$, the MRPs σ_i^e will be switched to its shadow counterpart $s(\sigma_i^e)$. On the other hand, if the shadow counterpart $s(\sigma_i^e)$ increases to $1 + \delta_i$, it will be switched to the MRPs σ_i^e . Compared with the switching surface $\|\sigma_i^e\| = 1$ of [40] and [43] that is sensitive to perturbation and measurements noise, the switching surface $\|\sigma_i^e\| = 1 + \delta_i$ uses the hysteresis switching technique, which has stronger robustness.

The adaptive update law of $\hat{\theta}_i$ is designed as

$$\dot{\hat{\theta}}_i = \mu_i \left(\Upsilon(\Lambda_i - \dot{\zeta}_i) - \omega_i^{\times} \Upsilon(\omega_i) \right)^T (\omega_i^e - \zeta_i) \quad (19)$$

with $\mu_i > 0$.

Remark 3: Note that the quaternion-based hybrid controller was first proposed in [38] to solve the unwinding problem, where the switching dynamics of a binary logic variable $h \in \{-1, 1\}$ is designed to adjust the attitude rotation orientation. In contrast to the results in [38], the advantages of MRPs-based hybrid control scheme (17) are threefold.

- 1) The proposed controller (17) does not need to add the binary logic variable and, therefore, is more easier to implement.
- 2) Two MRPs sets σ_i^e and $s(\sigma_i^e)$ own the same equilibrium while unit-quaternion has two different equilibriums. This nice property makes the global stability analysis of the MRPs-based controller (17) simpler than that of [38].
- 3) The controller of [38] relies on the precise inertia matrix while the proposed controller (17) does not.

D. Stability Analysis

In this section, we provide the stability analysis of the whole error systems (4), (8), and (9) under the proposed control force (12) and control torque (17). According to the stability analysis of hybrid control framework in [44], we first show that

the attitude and angular velocity errors (σ_i^e, ω_i^e) asymptotically converge to zero in the following proposition.

Proposition 1: Suppose that Assumption 1 holds. Consider the attitude error dynamical systems (8) and (9) of the i th UAV. The developed MRPs-based adaptive hybrid controller (17) guarantees that $\lim_{t \rightarrow \infty} \sigma_i^e(t) = 0$ and $\lim_{t \rightarrow \infty} \omega_i^e(t) = 0$ for any initial states $\sigma_i^e(0)$ and $\omega_i^e(0)$.

Proof: Define the variable $s_i = \omega_i^e - \dot{\zeta}_i$. From (16) and (17), it follows that:

$$\begin{aligned} J_i \dot{s}_i &= -\omega_i^\times \Upsilon(\omega_i) \theta_i + \tau_i + \Upsilon(\Lambda_i) \theta_i - J_i \dot{\zeta}_i \\ &= -k_\omega s_i - k_\sigma \sigma_i^e - (\Upsilon(\Lambda_i - \dot{\zeta}_i) - \omega_i^\times \Upsilon(\omega_i)) \tilde{\theta}_i \end{aligned} \quad (20)$$

where $\tilde{\theta}_i = \hat{\theta}_i - \theta_i$ is the estimation error.

Choose the following Lyapunov function:

$$\begin{aligned} V_1(x_i, \zeta_i, \tilde{\theta}_i) &= 2k_\sigma \ln(1 + \|\sigma_i^e\|^2) + \frac{1}{2} s_i^T J_i s_i \\ &\quad + \frac{1}{2l_2} \|\zeta_i\|^2 + \frac{1}{2\mu_i} \tilde{\theta}_i^T \tilde{\theta}_i. \end{aligned}$$

Obviously, $V_1(x_i, \zeta_i, \tilde{\theta}_i)$ is positive definite.

Based on (8) and (17)–(20), the derivative of $V_1(x_i, \zeta_i, \tilde{\theta}_i)$ along the flow set $\mathcal{C}_i$ is obtained as

$$\begin{aligned} \dot{V}_1(x_i, \zeta_i, \tilde{\theta}_i) &= 4k_\sigma \frac{(\sigma_i^e)^T}{1 + \|\sigma_i^e\|^2} E(\sigma_i^e) \omega_i^e - k_\omega s_i^T s_i \\ &\quad - k_\sigma (\omega_i^e - \zeta_i)^T \sigma_i^e - s_i^T (\Upsilon(\Lambda_i - \dot{\zeta}_i) \\ &\quad - \omega_i^\times \Upsilon(\omega_i)) \tilde{\theta}_i + \frac{1}{l_2} \zeta_i^T \dot{\zeta}_i + \frac{1}{\mu_i} \tilde{\theta}_i^T \dot{\tilde{\theta}}_i. \end{aligned}$$

Note from (2) that $(4(\sigma_i^e)^T / 1 + \|\sigma_i^e\|^2) E(\sigma_i^e) \omega_i^e = (\sigma_i^e)^T \omega_i^e$. It then follows that:

$$\begin{aligned} \dot{V}_1(x_i, \zeta_i, \tilde{\theta}_i) &= -k_\omega s_i^T s_i + \zeta_i^T \left(k_\sigma \sigma_i^e + \frac{1}{l_2} \dot{\zeta}_i \right) \\ &= -k_\omega s_i^T s_i - \frac{l_1}{l_2} \zeta_i^T \zeta_i \end{aligned} \quad (21)$$

where the last equality is derived by using (18). Integrating two sides of (21), we have that

$$V_1(t) + \int_0^t k_\omega s_i^T(\tau) s_i(\tau) d\tau + \frac{l_1}{l_2} \zeta_i^T(\tau) \zeta_i(\tau) d\tau \leq V_1(0).$$

This infers that $s_i, \zeta_i \in \mathbb{L}_2$, and $V_1(x_i, \zeta_i, \tilde{\theta}_i) \in \mathbb{L}_\infty$, which also implies $s_i, \zeta_i, \sigma_i^e, \tilde{\theta}_i \in \mathbb{L}_\infty$. Based on (18) and (23), we obtain that $\dot{s}_i$ and $\dot{\zeta}_i \in \mathbb{L}_\infty$ are bounded. According to the Barbalat lemma [45], it follows that $\lim_{t \rightarrow \infty} s_i(t) = 0$ and $\lim_{t \rightarrow \infty} \zeta_i(t) = 0$. Since $s_i = \omega_i^e - \dot{\zeta}_i$, one has that $\lim_{t \rightarrow \infty} \omega_i^e(t) = 0$.

We next prove that $\lim_{t \rightarrow \infty} \sigma_i^e(t) = 0$ holds. Taking the derivative of (18) yields $\dot{\zeta}_i = -l_1 \dot{\zeta}_i - k_\sigma l_2 \dot{\sigma}_i^e$. From $\sigma_i^e, \zeta_i \in \mathbb{L}_\infty$, it follows from (18) that $\dot{\zeta}_i \in \mathbb{L}_\infty$. In addition, $\dot{\sigma}_i^e = E(\sigma_i^e) \omega_i^e \in \mathbb{L}_\infty$ can be obtained from $\omega_i^e \in \mathbb{L}_\infty$ and $\|E(\sigma_i^e)\| \leq (1 + \|\sigma_i^e\|^2/4)$ shown in (2). Based on the above results, we obtain that $\dot{\zeta}_i \in \mathbb{L}_\infty$. By combining $\lim_{t \rightarrow \infty} \zeta_i(t) = 0$, one has that $\lim_{t \rightarrow \infty} \dot{\zeta}_i(t) = 0$ by using the Barbalat lemma. Since $\sigma_i^e = -(1/l_2 k_\sigma)(\dot{\zeta}_i + l_1 \zeta_i)$, it then follows that $\lim_{t \rightarrow \infty} \sigma_i^e(t) = 0$.

On the other hand, when the state x belongs to the jump set, that is, $x_i \in \mathcal{D}_i$, the change of $V_1(x_i, \zeta_i, \tilde{\theta}_i)$ is given by

$$\begin{aligned} &V_1(G(x_i), \zeta_i, \tilde{\theta}_i) - V_1(x_i, \zeta_i, \tilde{\theta}_i) \\ &= 2k_\sigma \ln \left(\frac{1 + \|\sigma_i^e\|^{-2}}{1 + \|\sigma_i^e\|^2} \right) = -4k_\sigma \ln(\|\sigma_i^e\|). \end{aligned}$$

Since $\|\sigma_i^e\| \geq 1 + \delta_i$ for the case that $x_i \in \mathcal{D}_i$, it then follows that:

$$V_1(G(x_i), \zeta_i, \tilde{\theta}_i) - V_1(x_i, \zeta_i, \tilde{\theta}_i) \leq -4k_\sigma \ln(1 + \delta_i) < 0.$$

This implies that $V_1(x_i, \zeta_i, \tilde{\theta}_i)$ is strictly decreasing in the jump set $\mathcal{D}_i$. By combining the above results in the flow set and jump set, we conclude that $\lim_{t \rightarrow \infty} \sigma_i^e(t) = 0$ and $\lim_{t \rightarrow \infty} \omega_i^e(t) = 0$ for any initial states $\sigma_i^e(0)$ and $\omega_i^e(0)$. ■

Based on the results of Proposition 1, we next prove that the developed control force (12) guarantees that the position and velocity errors of system (4) converge to zero.

Let $g = \text{col}(g_1, \dots, g_m)$ and $g^* = \text{col}(g_1^*, \dots, g_m^*)$ be the stack vectors of the bearing vectors $\{g_{ij}\}_{(i,j) \in \mathcal{E}}$ and the desired bearing vectors $\{g_{ij}^*\}_{(i,j) \in \mathcal{E}}$, respectively. One useful lemma is provided as follows.

Lemma 3 [34]: Suppose that no agents coincide in p^* or p , it holds that

$$p^T \bar{H}(g - g^*) \geq 0 \quad (22)$$

where the equality holds if and only if $g = g^*$.

Proposition 2: Consider the error dynamical system (4). Under Assumption 2 and the condition $\lim_{t \rightarrow \infty} \sigma_i^e(t) = 0$, the developed adaptive control force (12) guarantees that $\lim_{t \rightarrow \infty} p_i^e(t) = 0$ and $\lim_{t \rightarrow \infty} v_i^e(t) = 0$ for $\forall i \in \mathcal{V}_f$.

Proof: Substituting (12) into (4) yields

$$\begin{aligned} \dot{p}_i^e &= v_i^e, \\ m_i \dot{v}_i^e &= -m_i g e_3 - k_s \tanh(\varpi_i) \end{aligned} \quad (23a)$$

$$+ k_p \sum_{j \in \mathcal{N}_i} (g_{ij} - g_{ij}^*) + \hat{m}_i (g \hat{e}_3 + \dot{\chi}_i) + \varepsilon_i \quad (23b)$$

where $\varepsilon_i = T_i(R(\sigma_i) - R(\sigma_i^e))e_3$.

Let $r_i = \sum_{j \in \mathcal{N}_i} (g_{ij} - g_{ij}^*)$, $i \in \mathcal{V}_f$ and $r_f = \text{col}(r_1, \dots, r_{n_f})$. According to the definitions of $\bar{H}$, g , and g^* , it follows that:

$$r_f = -B \bar{H}^T (g - g^*) \quad (24)$$

where $B = [I_{3n_f}, 0_{3n_f \times 3n_f}] \in \mathbb{R}^{3n_f \times 3n_f}$.

Define $p_f^e = \text{col}(p_1^e, \dots, p_{n_f}^e)$, $v_f^e = \text{col}(v_1^e, \dots, v_{n_f}^e)$, $M = \text{diag}(m_1 I_3, \dots, m_{n_f} I_3)$, $\hat{M} = \text{diag}(\hat{m}_1 I_3, \dots, \hat{m}_{n_f} I_3)$, $\dot{\chi} = \text{col}(\dot{\chi}_1, \dots, \dot{\chi}_{n_f})$, $\chi = \text{col}(\chi_1, \dots, \chi_{n_f})$, $\varpi = \text{col}(\varpi_1, \dots, \varpi_{n_f}) = v_f - \chi$, and $\varepsilon = \text{col}(\varepsilon_1, \dots, \varepsilon_{n_f})$. The compact format of (23) is given by

$$\begin{aligned} \dot{p}_f^e &= v_f^e \\ M \dot{v}_f^e &= -M(\mathbf{1}_{n_f} \otimes g \hat{e}_3) - k_s \tanh(\varpi) \end{aligned} \quad (25a)$$

$$+ k_p r_f + \hat{M}((\mathbf{1}_{n_f} \otimes g \hat{e}_3) + \dot{\chi}) + \varepsilon. \quad (25b)$$

Similarly, the compact format of auxiliary system (13) is

$$\dot{\xi} = -k_1 \tanh(\xi - \chi) \quad (26a)$$

$$\dot{\chi} = -k_3 \dot{\xi} + k_2 k_p r_f = k_1 k_3 \tanh(\xi - \chi) + k_2 k_p r_f \quad (26b)$$

with $\xi = \text{col}(\xi_1, \dots, \xi_f)$.

Consider the following Lyapunov function:

$$V_2 = k_p p^T \bar{H}^T (g - g^*) + \frac{1}{2} \varpi^T M \varpi + \frac{1}{2k_2} \|\chi - \bar{v}_d\|^2 + \frac{k_3}{2k_2} \|\xi - \bar{v}_d\|^2 + \frac{1}{2} \sum_{i=1}^{n_f} \gamma_i^{-1} \|\tilde{m}_i\|^2$$

where $\tilde{m}_i = \hat{m}_i - m_i$, $i \in \mathcal{V}_f$ and $\bar{v}_d = \mathbf{1}_{n_f} \otimes v_d$. Based on Lemma 3, one has that V_2 is positive definite.

The derivative of V_2 along (25) and (26) is obtained as

$$\begin{aligned} \dot{V}_2 &= k_p p^T \bar{H}^T \dot{g} + k_p (g - g^*)^T \bar{H} \dot{p} + \varpi^T M \dot{\varpi} \\ &+ \frac{1}{k_2} (\chi - \bar{v}_d)^T \dot{\chi} + \frac{k_3}{k_2} (\xi - \bar{v}_d)^T \dot{\xi} + \sum_{i=1}^{n_f} \gamma_i^{-1} \tilde{m}_i \dot{\tilde{m}}_i. \end{aligned}$$

Note that $p^T \bar{H}^T \dot{g} = \sum_{k=1}^m e_k^T \dot{g}_k = \sum_{k=1}^m e_k^T P_{g_k} \cdot (\dot{e}_k / \|e_k\|) = 0$ since $P_{g_k} e_k = 0$. In addition, from $v_l = \mathbf{1}_{n_l} \otimes v_d$, one has that

$$\begin{aligned} (g - g^*)^T \bar{H} \dot{p} &= (g - g^*)^T \bar{H} \left(\begin{bmatrix} v_f - \mathbf{1}_{n_f} \otimes v_d \\ 0 \end{bmatrix} + \mathbf{1}_n \otimes v_d \right) \\ &= (g - g^*)^T \bar{H} \begin{bmatrix} I_{3n_f} \\ 0_{3n_l \times 3n_f} \end{bmatrix} (v_f - \mathbf{1}_{n_f} \otimes v_d) \\ &= (g - g^*)^T \bar{H} B^T (v_f - \bar{v}_d) \\ &= -r_f^T (v_f - \bar{v}_d) \end{aligned}$$

where the second equality is derived by using $\bar{H}(\mathbf{1}_n \otimes v_d) = 0$. According to the above results, we obtain that

$$\begin{aligned} \dot{V}_2 &= -k_p r_f^T (v_f - \bar{v}_d) + \varpi^T \left(-M(\mathbf{1}_{n_f} \otimes g \hat{e}_3 + \dot{\chi}) \right. \\ &\quad \left. - k_s \tanh(\varpi) + k_p r_f + \hat{M}((\mathbf{1}_{n_f} \otimes g \hat{e}_3) + \dot{\chi}) + \varepsilon \right) \\ &+ \frac{1}{k_2} (\chi - \bar{v}_d)^T \dot{\chi} + \frac{k_3}{k_2} (\xi - \bar{v}_d)^T \dot{\xi} + \sum_{i=1}^{n_f} \gamma_i^{-1} \tilde{m}_i \dot{\tilde{m}}_i \\ &= -k_s \varpi^T \tanh(\varpi) - k_p r_f^T v_f + k_p (v_f - \chi)^T r_f \\ &+ \bar{v}_d^T \left(k_p r_f - \frac{1}{k_2} \dot{\chi} - \frac{k_3}{k_2} \dot{\xi} \right) + \frac{1}{k_2} \chi^T \dot{\chi} + \frac{k_3}{k_2} \xi^T \dot{\xi} \\ &+ \sum_{i=1}^{n_f} \gamma_i^{-1} \tilde{m}_i \left(\dot{\hat{m}}_i + \gamma_i (g_3 \hat{e}_3 + \chi_i)^T \varpi_i \right). \end{aligned}$$

From the adaptive update law (14), one has that if $\dot{\hat{m}}_i = \gamma_i (g \hat{e}_3 + \dot{\chi}_i)^T \varpi_i$, then $\tilde{m}_i (\dot{\hat{m}}_i - \gamma_i (g_3 \hat{e}_3 + \chi_i)^T \varpi_i) = 0$; if $\dot{\hat{m}}_i = 0$, it then follows that:

$$\begin{aligned} \tilde{m}_i &\left(\dot{\hat{m}}_i + \gamma_i (g_3 \hat{e}_3 + \chi_i)^T \varpi_i \right) \\ &= -\gamma_i (m_i - \hat{m}_i) (g_3 \hat{e}_3 + \chi_i)^T \varpi_i \\ &= \gamma_i (m_i - \hat{m}_i) \psi_i \leq 0. \end{aligned}$$

Then, by combining (26), $\dot{V}_2$ can be simplified as

$$\dot{V}_2 \leq -k_s \varpi^T \tanh(\varpi) + \varpi^T \varepsilon + \chi^T \left(k_p r_f + \frac{1}{k_2} \dot{\chi} \right) + \frac{k_3}{k_2} \xi^T \dot{\xi}$$

$$\begin{aligned} &= -k_s \varpi^T \tanh(\varpi) + \varpi^T \varepsilon - \frac{k_3}{k_2} \chi^T \dot{\xi} + \frac{k_3}{k_2} \xi^T \dot{\xi} \\ &= -k_s \varpi^T \tanh(\varpi) + \varpi^T \varepsilon \\ &\quad - \frac{k_1 k_3}{k_2} (\xi - \chi)^T \tanh(\xi - \chi). \end{aligned} \quad (27)$$

From the definition of V_2 , one has that $V_2 \geq (1/2) \varpi^T M \varpi \geq (1/2) \min_{i \in \mathcal{V}_f} (m_i) \|\varpi\|^2$, which implies that $\|\varpi\| \leq \rho \sqrt{V_2}$ with $\rho = \sqrt{2 / \min_{i \in \mathcal{V}_f} (m_i)}$. In addition, we note that $\varepsilon_i = T_i(R(\sigma_i) - R(\sigma_i^e))e_3$ is bounded since T_i shown in (15) is bounded and $\|(R(\sigma_i) - R(\sigma_i^e))e_3\| \leq 2$. Moreover, from Proposition 1, it follows that $\lim_{t \rightarrow \infty} \sigma_i^e(t) = 0$ and further derive that $\lim_{t \rightarrow \infty} R(\sigma_i^e) = I_3$ for $\forall \in \mathcal{V}_f$. Since $\varepsilon_i = T_i R(\sigma_i^e) (R(\sigma_i^e) - I_3) e_3$, one has that ε_i asymptotically converges to the origin. According to [45], there exists a $\mathcal{KL}$ -class function β such that $\|\varepsilon(t)\| \leq \beta(\|\varepsilon(0)\|, t)$ for $\forall t \geq 0$. From the above analysis, one has that $\dot{V}_2 \leq \rho \sqrt{V_2} \beta(\|\varepsilon(0)\|, t)$. By using the comparison lemma and Massera Lemma of [45, Lemma C.1], we obtain that $\sqrt{V_2(t)} \leq \int_0^\infty \rho \beta(\|\varepsilon(0)\|, \tau) d\tau + \sqrt{V_2(0)} \leq \vartheta_m$, where ϑ_m is a bounded constant. It then follows from (27) that:

$$\begin{aligned} \dot{V}_2 &\leq -k_s \varpi^T \tanh(\varpi) - \frac{k_1 k_3}{k_2} (\xi - \chi)^T \tanh(\xi - \chi) \\ &\quad + \rho \vartheta_m \beta(\|\varepsilon(0)\|, t). \end{aligned} \quad (28)$$

Integrating both sides of (28), based on the comparison lemma and Massera Lemma of [45], one has that

$$\begin{aligned} V_2(t) &+ k_s \int_0^t \varpi^T(\tau) \tanh(\varpi(\tau)) d\tau \\ &+ \frac{k_1 k_3}{k_2} \int_0^t (\xi(\tau) - \chi(\tau))^T \tanh(\xi(\tau) - \chi(\tau)) d\tau \\ &\leq \int_0^t \rho \vartheta_m \beta(\|\varepsilon(0)\|, \tau) d\tau + V_2(0) < \infty. \end{aligned} \quad (29)$$

This implies that $\varpi, \xi - v_d, \chi - v_d \in \mathbb{L}_\infty$, and $\varpi, \xi - \chi \in \mathbb{L}_2$. Since v_d is bounded, it then follows that ξ and χ are both bounded. In addition, according to (1) and (26), one further can derive that $\dot{\varpi}$ and $\dot{\xi} - \dot{\chi}$ are both bounded. By using the Babarlat lemma, we obtain that $\lim_{t \rightarrow \infty} \varpi(t) = 0$ and $\lim_{t \rightarrow \infty} (\xi(t) - \chi(t)) = 0$. From (26), we further obtain that

$$\ddot{\xi} - \ddot{\chi} = -\frac{(1+k_3)k_1}{\cosh^2(\xi - \chi)} (\dot{\xi} - \dot{\chi}) - k_2 k_p \dot{r}_f \quad (30)$$

where $\dot{r}_f = B \bar{H}^T \dot{g} = B \bar{H}^T \text{diag}((P_{g_1}/\|e_1\|), \dots, (P_{g_m}/\|e_m\|)) \bar{H} v$. Since ϖ and χ are both bounded, it then follows that $v_f = \varpi + \chi \in \mathbb{L}_\infty$ and $v \in \mathbb{L}_\infty$. In addition, suppose that no collisions happen among the agents, one has that $\|e_k\| \geq d_c$ for $\forall k \in \mathcal{E}$, where d_c is a bounded constant. Then, we obtain that $\dot{r}_f$ is bounded. Since $\cosh^2(\xi - \chi) \geq 1$ and $\dot{\xi} - \dot{\chi}$ is bounded, we obtain $\ddot{\xi} - \ddot{\chi} \in \mathbb{L}_\infty$. Based on the Babarlat lemma, $\lim_{t \rightarrow \infty} (\dot{\xi}(t) - \dot{\chi}(t)) = 0$. From $\dot{\xi} = -k_1 \tanh(\xi - \chi)$, one has that $\lim_{t \rightarrow \infty} \dot{\xi}(t) = 0$ and $\lim_{t \rightarrow \infty} \dot{\chi}(t) = 0$. In addition, from $\dot{\chi} = -k_3 \dot{\xi} - k_2 k_p r_f$, it follows that $\lim_{t \rightarrow \infty} r_f(t) = 0$, which implies $\lim_{t \rightarrow \infty} p^T(t) \bar{H}^T (g(t) - g^*) = 0$. According to Lemma 3, one has that $\lim_{t \rightarrow \infty} (g(t) - g^*) = 0$. Under Assumption 2, we obtain that the target formation $(\mathcal{G}, p^*)$ is achieved, that is, $\lim_{t \rightarrow \infty} p_i^e(t) = 0$ and $\lim_{t \rightarrow \infty} v_i^e(t) = 0$ for $\forall i \in \mathcal{V}_f$. ■

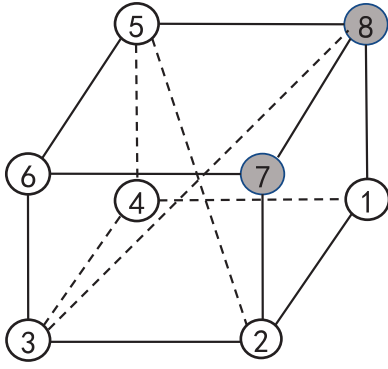


Fig. 1. Network topology of target formation $(\mathcal{G}, p^*)$.

Based on Propositions 1 and 2, we present the main results of this article as follows.

Theorem 1: Consider multiple VTOL UAVs modeled by system (1) subject to unknown physical parameters. Under Assumptions 1 and 2, the proposed adaptive control force (12) and MRPs-based hybrid control torque (17) achieve the formation control objective (3).

Proof: It is shown in Proposition 1 that $\lim_{t \rightarrow \infty} \sigma_i^e(t) = 0$ and $\lim_{t \rightarrow \infty} \omega_i^e(t) = 0$ for any initial states $\sigma_i^e(0)$ and $\omega_i(0)$. In addition, Proposition 2 has shown that $\lim_{t \rightarrow \infty} p_i^e(t) = 0$ and $\lim_{t \rightarrow \infty} v_i^e(t) = 0$. According to Lemma 1, we obtain that the formation control objective (3) is achieved. ■

Remark 4: Compared with the MRPs-based hybrid attitude controller in [46] that relies on the accurate information of the inertia matrix, the attitude tracking controller (17) is synthesized by combining hybrid method and adaptive technique, which does not rely on the precise model. In addition, from the proofs of Proposition 1, we know that the auxiliary variable ζ_i is crucial in the convergence analysis, which avoids causing the cross term of [25] that will result in a complex switching condition in the jump set $\mathcal{D}_i$.

IV. SIMULATION RESULTS

In this section, we carry out a numerical example to verify the effectiveness of the developed adaptive formation control algorithm for multiple VTOL UAVs with unknown physical parameters. A formation of eight heterogeneous VTOL UAVs is considered, which includes six followers (UAVs 1–6) and two leaders (UAVs 7 and 8). The target formation $(\mathcal{G}, p^*)$ shown in Fig. 1 is a cube in the 3-D space, which is infinitesimally bearing rigid and its bearing constraints $\{g_{ij}^*\}_{(i,j) \in \mathcal{E}}$ are $g_{12}^* = g_{43}^* = g_{56}^* = g_{87}^* = [1, 0, 0]^T$, $g_{41}^* = g_{32}^* = g_{67}^* = g_{58}^* = [0, 1, 0]^T$, $g_{18}^* = g_{27}^* = g_{36}^* = g_{45}^* = [0, 0, 1]^T$, $g_{38}^* = [-(1/\sqrt{3}), (1/\sqrt{3}), (1/\sqrt{3})]^T$, and $g_{25}^* = [-(1/\sqrt{3}), -(1/\sqrt{3}), (1/\sqrt{3})]^T$. As shown in Section II, under the prescribed bearing constraints $\{g_{ij}^*\}_{(i,j) \in \mathcal{E}}$, the motion of whole target formation $(\mathcal{G}, p^*)$ is dominated by these two leaders. We set the velocities of the leaders to be a common constant $v_d = [0.2, 0.1, 0.2]^T$ and their initial positions are $p_3(0) = [10, 10, 10]^T$ and $p_4(0) = [0, 10, 10]^T$.

The physical parameters including mass and inertia matrix of all follower UAVs are presented in Table I. The states $(p_i, v_i, \sigma_i, \omega_i)$ of the followers are initialized as $p_1(0) = [1, 10, 1]^T$, $p_2 = [12, 2, 1]^T$, $p_3(0) = [8, 0, -5]^T$, $p_4(0) =$

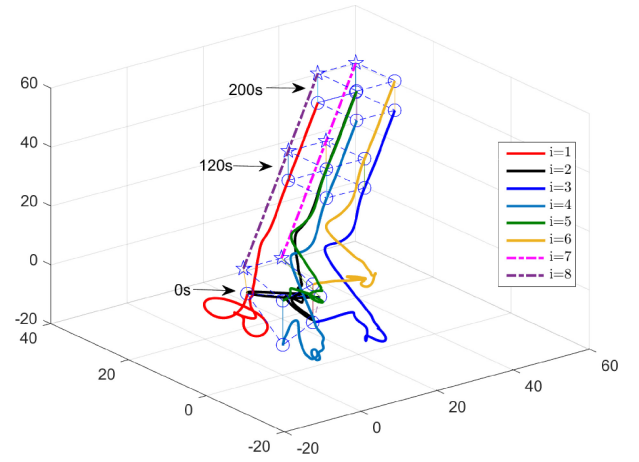


Fig. 2. Whole UAV formation motion.

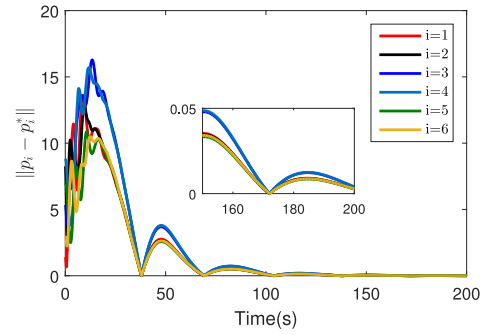


Fig. 3. Position errors $\|p_i - p_{f,i}^*\|$.

$[-5, -5, -5]^T$, $p_5(0) = [-5, -5, 10]^T$, $p_6(0) = [8, 0, 8]^T$, $\sigma_1(0) = [4, -2, 1]^T$, $\sigma_2(0) = [-1, 2, 2]^T$, $\sigma_3(0) = [1, 0, 1]^T$, $\sigma_4(0) = [0, 1, -1]^T$, $\sigma_5(0) = [-0.8, 0, 0.5]^T$, $\sigma_6(0) = [-0.2, 0.5, 0.1]^T$, $v_i = [0, 0, 0]^T$ and $\omega_i(0) = [0, 0, 0]^T$, $i = 1, 2, \dots, 6$. The gain parameters of the distributed control force u_i in (12)–(14) are selected as $k_s = 1$, $k_p = 1$, $k_1 = 0.1$, $k_2 = 4$, $k_3 = 4$, $\gamma_i = 1$, $i = 1, 2, \dots, 6$, and the gain parameters of attitude control torque τ_i in (17) and (18) are $k_\omega = 0.2$, $k_\sigma = 0.2$, $l_1 = 1$, $l_2 = 2$ and $\delta_i = 0.2$. The initial values of auxiliary variables ξ_i , χ_i , ζ_i in (13) and (18) are chosen from a uniform distribution over the interval $(0, 1)$. The estimation variables $\hat{m}_i$, $\hat{\theta}_i$ are initialized as $\hat{m}_i(0) = 1$ kg, $i = 1, 2, \dots, 6$, $\hat{\theta}_1(0) = [0.12, 0.12, 0.3, 0.01, 0.03, 0.02]^T$ kg·m² and $\hat{\theta}_2(0) = [0.04, 0.04, 0.08, 0.002, 0.003, 0.004]^T$ kg·m².

Fig. 2 depicts the whole formation motion in the 3-D space and formation configurations at the times 0s, 120s and 200s. Figs. 3 and 4 exhibit the position error $\|p_i - p_{f,i}^*\|$ and velocity error $\|v_i - v_{f,i}^*\|$, $i = 1, 2, \dots, 6$. From Figs. 2–4, we observe that the proposed distributed control force (12) achieves the formation control objective in the sense of (3). We show the online estimation of the unknown mass parameter m_i in the left subfigure of Fig. 5 and the required thrust magnitude T_i in the right subfigure.

In addition, we further demonstrate that the proposed HAC law (17) achieves the command attitude tracking and avoids the unwinding problem simultaneously. Without loss of generality, we only show the attitude tracking results for the first two followers UAVs 1–2. Based on the results of

TABLE I
PHYSICAL PARAMETERS OF UAVs

UAV	Mass m_i (kg)	Inertia matrix J_i ($\text{kg} \cdot \text{m}^2$)
UAV 1	1.2	$[1.522, 0.22, 0.33; 0.22, 1.522, 0.11; 0.33, 0.11, 3] \times 10^{-1}$
UAV 2	0.85	$[4.856, 0.44, 0.33; 0.44, 4.856, 0.22; 0.33, 0.22, 8.801] \times 10^{-2}$
UAV 3	0.35	$[2.56, 0.11, 0.22; 0.11, 2.56, 0.22; 0.22, 0.22, 5.2] \times 10^{-2}$
UAV 4	3.4	$[3.54, 0.33, 0.11; 0.33, 3.54, 0.22; 0.11, 0.22, 6.54] \times 10^{-1}$
UAV 5	1.0	$[1.1, 0.11, 0.11; 0.11, 1.1, 0.22; 0.11, 0.22, 2.8] \times 10^{-1}$
UAV 6	2.0	$[2.2, 0.33, 0.44; 0.33, 2.2, 0.33; 0.44, 0.33, 4.8] \times 10^{-1}$

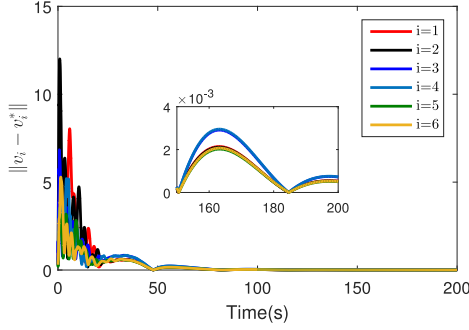


Fig. 4. Velocity errors $\|v_i - v_d\|$.

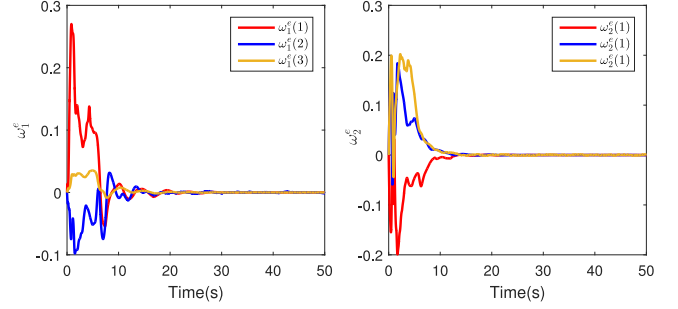


Fig. 7. Angular velocity error $\omega_i^e, i = 1, 2$.

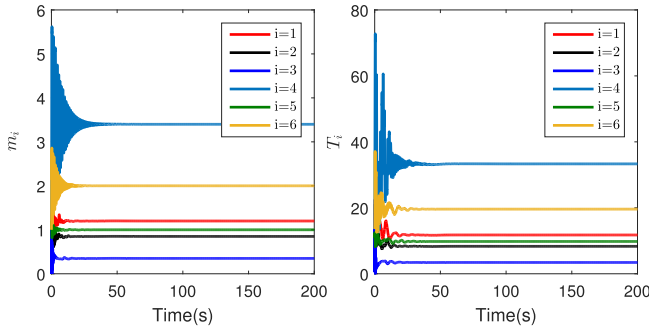


Fig. 5. Mass parameter estimation $\hat{m}_i$ and thrust magnitude T_i .

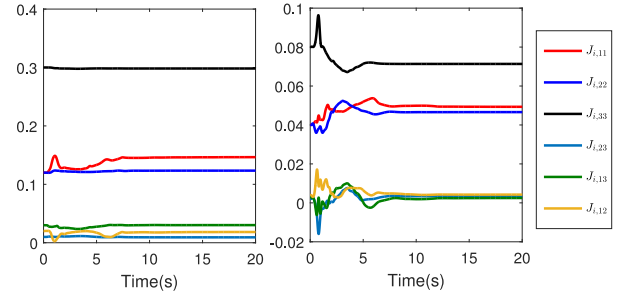


Fig. 8. Estimation of inertia matrix $J_i, i = 1, 2$.

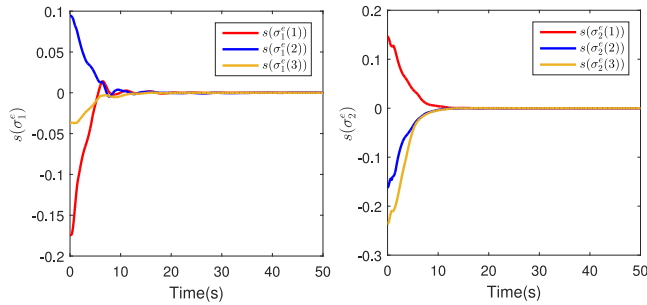


Fig. 6. Attitude errors $s(\sigma_i^e), i = 1, 2$.

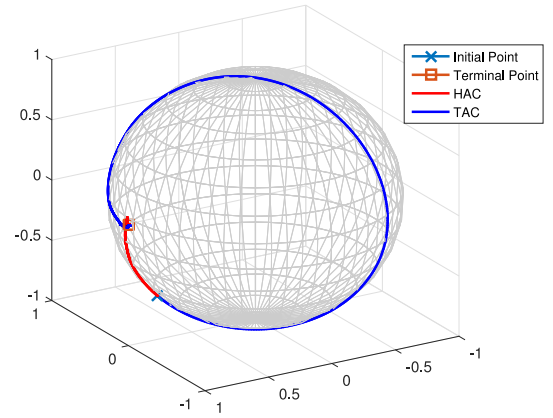


Fig. 9. Changes of rotation angle in terms of $s(\sigma_i^e)$ under two control laws.

Lemma 1, we extract the initial values of command attitude of two followers as $\sigma_1^c(0) = [0.0121, -0.0172, 0]^T$ and $\sigma_2^c(0) = [-0.0629, -0.0216, 0]^T$, and then obtain the initial relative attitudes $\sigma_1^e(0) = [4.2746, -2.3189, 0.9104]^T$ and $\sigma_2^e(0) = [-1.4124, 1.5668, 2.2860]^T$. Since $\|\sigma_1^e(0)\| = 4.9476 > 1 + \delta_1 = 1.2$ and $\|\sigma_2^e(0)\| = 3.1105 > 1.2$, it follows from (17) that the MRPs $\sigma_i^e, i = 1, 2$ will be switched to its shadow counterpart $s(\sigma_i^e) = -\sigma_i^e / ((\sigma_i^e)^T \sigma_i^e)$ at the initial time such that the unwinding problem can be avoided.

Figs. 6 and 7 depict the response curves of the attitude errors in terms of the shadow counterpart $s(\sigma_i^e)$ and angular velocity errors ω_i^e for all $i \in \mathcal{V}_f$. Fig. 8 shows the online estimation results of the unknown inertia matrix. From Figs. 6 and 7, we observe that the attitude errors of all the followers converge to zero. In addition, to show the advantages of the developed HAC law (17), we compare it with the traditional attitude controller (TAC) given in [41] that did not consider the unwinding problem of MRPs. Figs. 9 and 10 show the

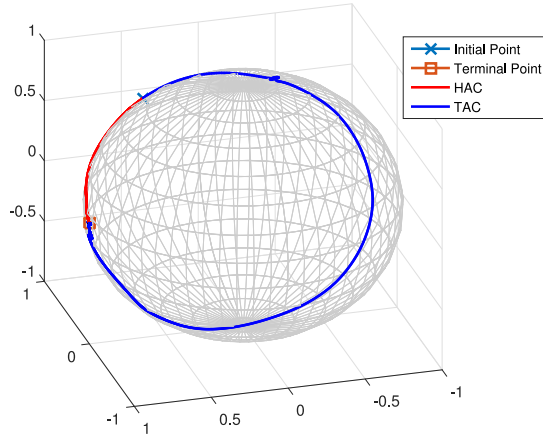


Fig. 10. Changes of rotation angle in terms of $s(\sigma_2^e)$ under two control laws.

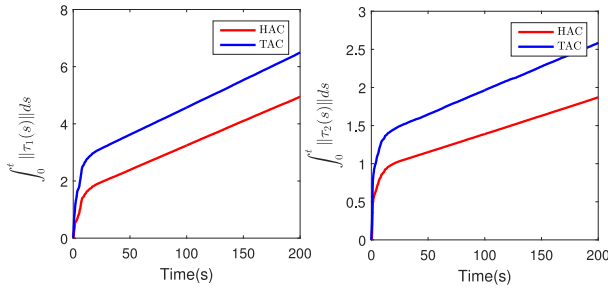


Fig. 11. Energy consumptions $E_i = \int_0^T \|\tau_i(s)\| ds$ of two control laws.

projections of attitude tracking errors $s(\sigma_i^e)$ of two followers on the unit sphere, which essentially describe the change of the rotation angle in terms of the MRPs $s(\sigma_i^e)$, $i = 1, 2$. From Figs. 9 and 10, we obtain that the proposed HAC law (17) guarantees that the command attitude can be tracked by rotating the small angles than those of the TAC method in [41]. It is illustrated that the proposed HAC law (17) effectively avoids the unwinding problem of MRPs. Moreover, we also compare the control energy consumptions $E_i = \int_0^T \|\tau_i(s)\| ds$ of these two attitude control laws in Fig. 11. It is shown that the proposed HAC law (17) requires less control consumption than that of TAC method in [41].

V. CONCLUSION

This article investigates the leader-following formation tracking control problem of multiple VTOL UAVs subject to parameter uncertainties. Under the hierarchical control framework, we first propose an adaptive formation control algorithm, which involves the bearing-only measurements with respect to the neighbors, to achieve no error tracking of target formation defined by interagent bearing vectors. Then, an MRPs-based adaptive hybrid control algorithm is developed, which achieves the command attitude tracking globally and avoids the unwinding problem simultaneously.

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